

Does Context-Length Order Matter in Training?

How I arrived at this study - three iterations of one question

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How I arrived at this study

I investigated one question: **Does context-length order matter in training?** For example, does curriculum-order training outperform alternative schedules? I worked through three iterations of this question.

I began with the hypothesis that ascending context training (32-64-128-256) would improve final long-context performance because the model could first learn short dependencies and then progressively extend its usable context. If that intuition were correct, curriculum training should outperform descending or unordered schedules, and the advantage should persist after the schedules ended.

The first exploratory result was surprisingly large. That forced me to examine whether the result came from training dynamics or from weaknesses in the measurement and experimental design. After correcting those problems, the second iteration showed almost no curriculum advantage over the baselines, but a clear anti-curriculum disadvantage. That appeared to support my original hypothesis. Only after I began looking for the strongest alternative explanation did I recognize that the most recent training context could explain the same endpoint pattern. The third iteration was designed to separate that recent-context explanation from a persistent ordering effect.

1. Iteration 1 - a result too dramatic to trust

The exploratory run produced large differences in the training curves and diagnostics. Instead of treating the dramatic result as a conclusion, I audited what the experiment was actually measuring.

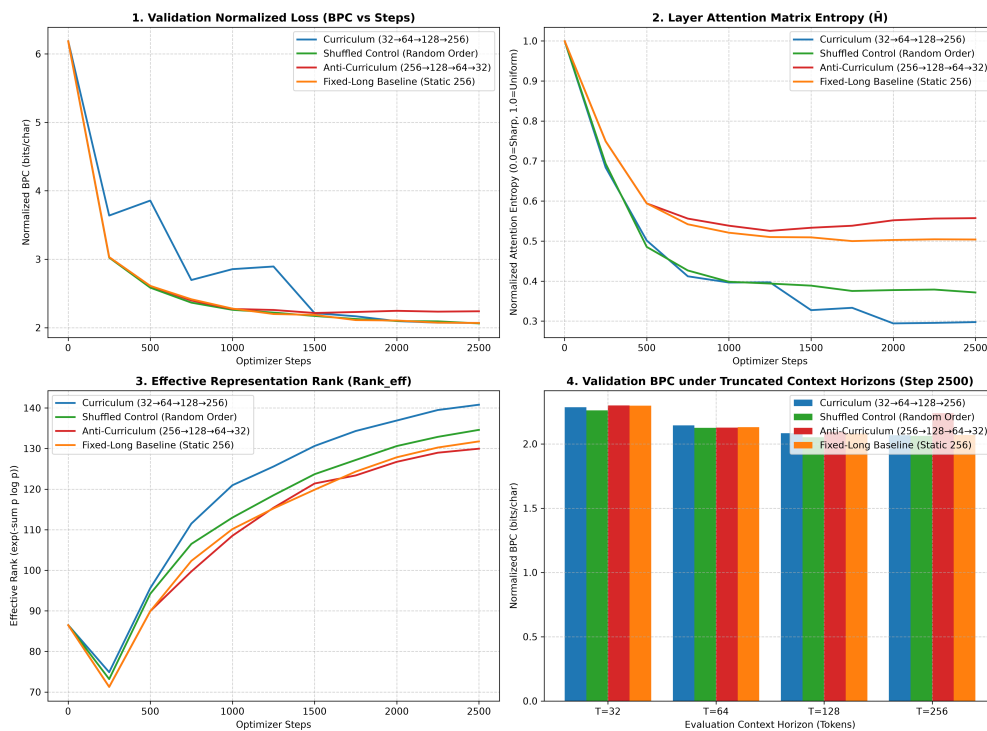


Figure 1 | Iteration 1: the question appeared important, but the evidence was exploratory. The pilot suggested schedule-dependent differences. Later checks found that one seed, unpaired batches, target-alignment problems, and overinterpreted diagnostics prevented a causal conclusion.

What failed: measurement, design, and interpretation

The first iteration used one seed; batches were not paired across schedules; different context horizons did not always score the same target positions; and a corpus probe suffered from resubstitution bias. I had also treated attention entropy and effective-rank differences as mechanism evidence when they were only exploratory correlations.

This distinction mattered: some problems were **measurement problems**, some were **experimental-design problems**, and some were problems in my **hypothesis and interpretation**. Fixing only one category would not have been enough.

2. Iteration 2 - a controlled paired pilot

I introduced five paired seeds, identical initialization within each seed, shared context-specific data manifests, held-out validation text, fixed target positions, and same-target evaluation. Curriculum showed almost no advantage over shuffled or fixed-long baselines, while anti-curriculum ended approximately 0.178 BPC worse at T=256 across the paired runs.



Figure 2 | Iteration 2: the anti-curriculum disadvantage replicated after measurement corrections. Pairing made the terminal comparison credible, but did not yet identify why it occurred. The original dashboard is shown as historical evidence; some of its mechanism language was later judged too strong.

Why Iteration 2 still could not answer the question

Ascending training ended at T=256, anti-curriculum ended at T=32, and the main evaluation also used T=256. The observed performance gap could therefore come from persistent temporal order, recent-context adaptation, or an interaction with optimizer state and the late learning-rate phase.

More seeds made the terminal comparison more reliable, but they could not repair a confounded estimand. I needed a new intervention, not merely a larger sample of the same design.

3. Iteration 3 - testing the strongest alternative explanation

To test whether recent context could explain the result, I added 500 identical T=256 recovery steps after the scheduled phase. I evaluated all arms at T=32, 64, 128, and 256 throughout recovery, using a 16-sequence process panel at 17 checkpoints and a nested 32-sequence anchor panel for the formal endpoints.

What was held constant

Within each paired seed, the model architecture, initialization, tokenizer, corpus split, exposure to every context length, 4,096 target tokens per update, scheduled data draws, recovery data draws, optimizer configuration, preserved optimizer state, global learning-rate schedule, training budget, validation sequences, target positions, and scoring code were fixed. Only the temporal permutation of the four pre-recovery context blocks varied in the primary comparison.

The estimands, thresholds, checkpoints, plots, and decision rules were preregistered before the formal results were inspected. One nonmonotonic schedule was retained as an exploratory control and was not allowed to redefine the primary conclusion.

The result contradicted my original judgment

Before recovery, the paired anti-curriculum-minus-curriculum T=256 difference was **+0.7570 BPC** (95% paired t interval [+0.3290, +1.1851]; 3/3 seeds positive). After 500 common T=256 steps, it was **-0.0485 BPC** (95% interval [-0.1491, +0.0521]; 0/3 positive). The process estimate fell from +0.8097 at recovery step 0 to +0.0719 after 10 steps and crossed zero by 50 steps.

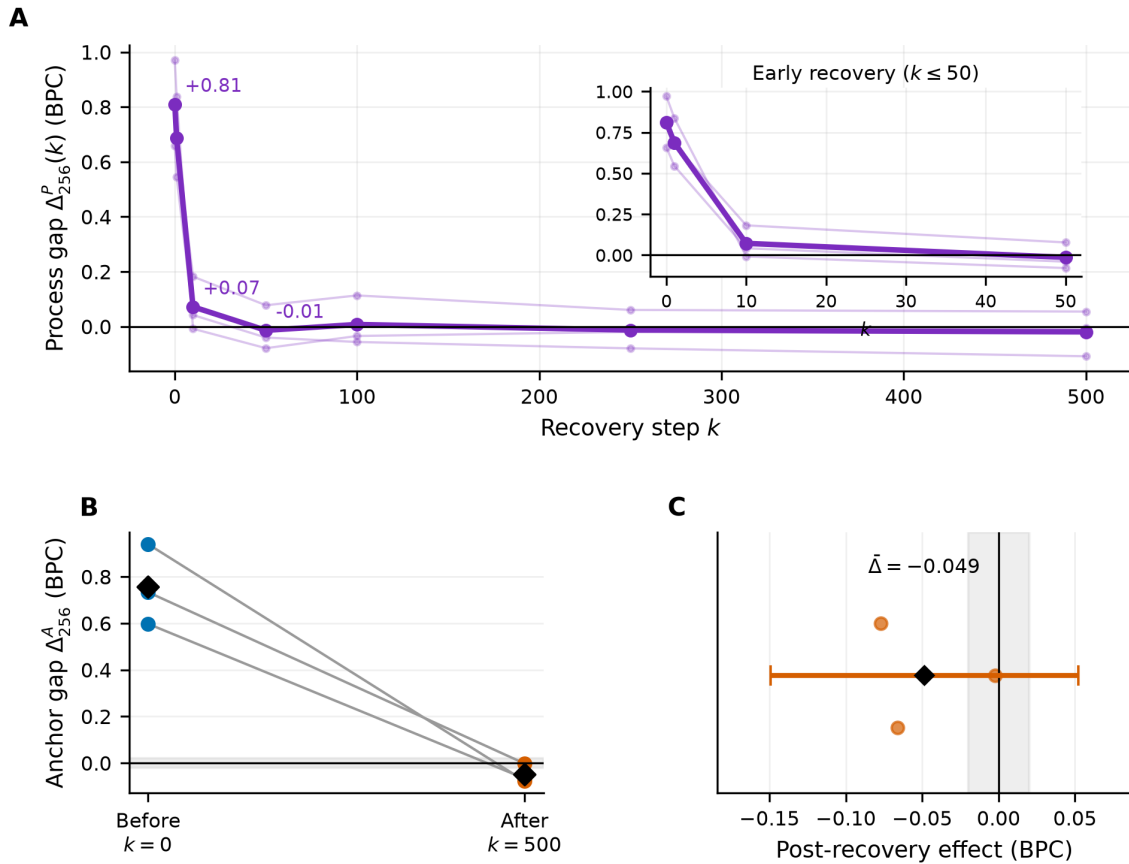


Figure 3 | Iteration 3: the large anti-curriculum performance gap rapidly reversed under common long-context recovery. Panel A shows the recovery trajectory, Panel B the three paired-seed endpoints, and Panel C the uncertain post-recovery mean. The smaller sign reversal remains outside the preregistered ± 0.02 BPC practical-removal band and is therefore unresolved.

4. Interpretation

My first explanation was that temporal order had placed anti-curriculum models in a persistently worse solution. That explanation became less plausible because the large $T=256$ difference disappeared within a small number of matched recovery steps. Before recovery, the difference was also concentrated at $T=256$ rather than shared across all evaluation horizons.

The reversal came mainly from improvement in the anti-curriculum arm: its mean $T=256$ BPC improved from 2.9980 to 2.2345, while the curriculum mean changed from 2.2410 to 2.2830. The finding was therefore not produced simply by the reference model becoming worse.

My original hypothesis was rejected in its strong form. The evidence supports recent-context adaptation more strongly than a large persistent ordering effect, although the experiment does not uniquely prove a recency mechanism.

5. Conclusion and remaining unknowns

The large anti-curriculum disadvantage did not persist after all models received the same long-context training condition. The result changed my interpretation from 'curriculum order creates a persistent advantage' to 'the most recent training context explains most of the observed final difference under this setup.' A smaller ordering effect cannot yet be ruled out.

The main failures were scientifically useful: Iteration 1 exposed measurement and pairing problems; Iteration 2 showed that reproducibility does not remove a confound; and an initial post-result attempt to classify every negative residual as practical recovery was withdrawn because that rule had not been preregistered.

Remaining unknowns

- whether the approximately -0.05 BPC post-recovery difference replicates with greater precision;
- which part of recovery - recent context, additional optimization, optimizer moments, or representation adaptation - caused the rapid change;
- whether optimizer-state resets would alter the trajectory;
- whether the result generalizes to other model sizes, tokenizers, corpora, context ranges, and schedules.

Across the three iterations, the question became narrower and the evidence became more discriminating. Corrected measurement established that the endpoint difference was reproducible, but not what caused it. The recovery intervention then showed that the large endpoint difference was not persistent. The remaining uncertainty is smaller and more specific: whether the reversed post-recovery difference is real, and which component of recovery produced the rapid change.

The main outcome is not that my first hypothesis survived. It is that the same question became better specified, the comparison became more controlled, and the remaining uncertainty became narrower and more honest.

Code, preregistration, full technical report, and formal artifacts: github.com/angelazu-builder/nanoGPT